Supplement to: When Does Score Diffusion Help Graph Conformal Prediction?

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This supplement contains the full proofs and secondary theory behind the main paper, followed by supporting diagnostics from the same stored experiments. References prefixed by "main paper" refer to the separately compiled manuscript.

S1 Proofs of the main results

We use the balanced CSBM and score assumptions of main-paper Definition 1 and Assumption 1. All moments condition on the focal node's label and degree $d \geq 1$.

S1.1 Neighbour signal (main-paper Lemma 1)

Proof. A neighbour w has $y_w = y_v$ with probability h, which gives $\mathbb{E}[s(w, y_v)] = m + (1-h)\Delta$. For $k \neq y_v$ it has $y_w = k$ with probability $(1-h)/(K-1)$, hence $\mathbb{E}[s(w,k)] = m + \Delta\big(1 - \frac{1-h}{K-1}\big)$. The difference is $\Delta\big(h - \frac{1-h}{K-1}\big) = \Delta\,\frac{Kh-1}{K-1}$. □

S1.2 Separation and optimal weight (main-paper Proposition 1)

Proof. Δ_λ follows from Lemma 1 and the own-score separation Δ. The own score is independent of the neighbours' scores, and by the law of total variance each $s(w, y_v)$ has variance $\tau^2 + h(1-h)\Delta^2$, and therefore $\mathrm{Var}[\bar{s}(v, y_v)] = \tau^2\rho^2/d$. Write $a = h_{\mathrm{adj}}$, $b = 1 - a$ and $c = \rho^2/d$. For $\Delta_\lambda > 0$ and $\lambda > 0$, $r_\lambda^2 > r_0^2$ is equivalent to $(1-\lambda b)^2 > (1-\lambda)^2 + \lambda^2 c$, which expands to $2\lambda a > \lambda^2(1 + c - b^2)$ and gives (6). Where $\Delta_\lambda > 0$, the derivative of r_λ^2 has the sign of $a - \lambda(a + c)$. For $a > 0$ the maximum is at $\lambda = a/(a+c)$, where $\Delta_\lambda > 0$. For $a \leq 0$ no $\lambda > 0$ improves the ratio (the non-positive-homophily result proved below), hence $\lambda^\star = 0$. □

S1.3 Non-positive homophily (main-paper Corollary 1)

Proof. If $\Delta_\lambda \leq 0$ the claim is immediate. Otherwise write $a = -\beta$ with $\beta \geq 0$. Then (6) becomes $\beta(\beta + 2 - 2/\lambda) > \rho^2/d$. Now $\Delta_\lambda > 0$ means $\lambda(1+\beta) < 1$, which implies $\beta + 2 - 2/\lambda < -\beta \leq 0$, and the left side is non-positive while $\rho^2/d > 0$. □

S1.4 Valid selectors (main-paper Proposition 2)

Proof. Condition on the graph, the features, the fitted classifier, the training and validation labels and the APS randomisation. The diffused true-label scores of all pool nodes are then fixed numbers, because λ and the diffusion use no calibration or test label. Since the calibration/test split is uniform, $\mathcal{V}_{\text{cal}} \cup \{u\}$ is an exchangeable draw from the pool, and the split-conformal argument applies, as in main-paper Sect. 5. □

S2 Degree threshold for efficiency

Corollary 1 (Degree and score efficiency). *If $h_{\text{adj}} > 0$, $\lambda^\star$ increases with d and tends to one. For fixed $\lambda > 0$ with $h_{\text{adj}} + 2/\lambda - 2 > 0$, condition (6) fails for all degrees at or below $d^\dagger = \rho^2/\big(h_{\text{adj}}(h_{\text{adj}} + 2/\lambda - 2)\big)$. A single global λ therefore over-diffuses low-degree nodes. If the denominator is non-positive, no degree satisfies the improvement condition.*

Improvement requires $d > d^\dagger$, holding h, Δ, τ fixed.

Proof. Write $a = h_{\text{adj}} > 0$ and $c = \rho^2/d$. The improvement condition is $a(a + 2/\lambda - 2) > c$. If its left side is non-positive, it cannot exceed $c > 0$. Otherwise solving for d yields $d > d^\dagger$. Moreover $\lambda^\star = ad/(ad+\rho^2)$ has derivative $a\rho^2/(ad+\rho^2)^2 > 0$ and limit one. □

S3 From separation to Gaussian set size

Let $p = (1-h)/(K-1)$ be the probability that a neighbour carries a particular wrong label $k \neq y_v$. Total variance gives

$$\tau^2_{\lambda,k} = \tau^2 \left[(1-\lambda)^2 + \frac{\lambda^2 \rho_k^2}{d} \right], \qquad \rho_k^2 = 1 + p(1-p)\frac{\Delta^2}{\tau^2}. \tag{S1}$$

Approximate the diffused true and wrong scores by Gaussians with means μ_t and $\mu_t + \Delta_\lambda$ and variances τ^2_λ and $\tau^2_{\lambda,k}$, respectively. At the population true-score quantile $q = \mu_t + z_{1-\alpha}\tau_\lambda$,

$$\mathbb{E}|\mathcal{C}(v)| \approx 1 - \alpha + (K-1)\Phi\left(\frac{z_{1-\alpha}\tau_\lambda - \Delta_\lambda}{\tau_{\lambda,k}} \right), \tag{S2}$$

where Φ is the standard normal CDF and $z_{1-\alpha}$ its quantile. Linearity of expectation suffices. Candidate-label inclusion events need not be independent. For $K = 2$, $p = 1 - h$ and both variances agree, so expected size decreases with $r_\lambda = \Delta_\lambda/\tau_\lambda$. For $K > 2$, the two variances differ and SNR alone does not determine expected size. Bounded APS scores need not be Gaussian, nor is the empirical finite-calibration quantile the population quantile. This is an interpretive approximation, not another finite-sample guarantee.

Table S1. Coverage at nominal 0.95 (GCN, 50 runs). Weighted and Mondrian columns apply to SNR-λ. Uniform rows are reference splits. Arrows identify the calibration and test populations.

		Uncorrected			SNR-λ corrected	
Graph	Calibration → test	APS	DAPS	SNR-λ	weighted	Mondrian
Computers	low→high	.969	.984	.985	.954	.967
Computers	high→low	.956	.941	.942	.957	.964
Computers	uniform (reference)	.949	.950	.950	.951	.955
Amazon-ratings	low→high	.951	.954	.954	.952	.956
Amazon-ratings	high→low	.951	.945	.944	.950	.952
Amazon-ratings	uniform (reference)	.949	.949	.950	.950	.953
ogbn-arxiv	$\leq$2017 → $\geq$2019	.969	.958	.950	.949	.953
ogbn-arxiv	uniform (reference)	.949	.949	.949	.950	.953

S4 Correlated-score moment extension

Retain the conditional means and marginal variances of the main-paper score model. Let distinct neighbour scores for the true candidate label have correlation r, and each node-to-neighbour pair have correlation κ, conditional on the focal label and degree. Thus own variance is τ^2, neighbour variance $\tau^2\rho^2$, and node-to-neighbour covariance $\kappa\tau^2\rho$. For a feasible covariance matrix,

$$\mathrm{Var}(\bar{s}) = \tau^2\rho^2\{1 + (d-1)r\}/d, \tag{S3}$$

$$\mathrm{Cov}(s, \bar{s}) = \kappa\tau^2\rho, \tag{S4}$$

$$\mathrm{Var}(\tilde{s}) = \tau^2\left[(1-\lambda)^2 + \lambda^2\rho^2\frac{1 + (d-1)r}{d} + 2\lambda(1-\lambda)\kappa\rho\right]. \tag{S5}$$

For $d > 1$, feasibility requires $-1/(d-1) \leq r \leq 1$ and $\kappa^2 \leq \{1+(d-1)r\}/d$. For $d = 1$, $|\kappa| \leq 1$. Setting $r = \kappa = 0$ recovers main-paper Eq. (5). Positive r leaves a neighbour-average noise floor as d grows, so the independent-score optimum need not approach one. These moment parameters have not been estimated for our GNNs. The extension motivates a limitation, not a validated replacement selector.

S5 Distribution shift diagnostic

Table S1 completes the comparison discussed in main-paper Sect. 7.4, including reverse degree shifts and Mondrian calibration. On Computers, reversing the shift changes SNR-λ coverage from 0.942 to 0.985. Mondrian calibration is conservative in these runs. The table allows direct comparison with propensity weighting for each shift direction.

S6 Agreement-based selector specifications

The following rules supplement the comparison in main-paper Sect. 5. Here $\mathrm{clip}(x, 0, 1)$ truncates x to $[0, 1]$.

Label-free, node-wise. HEAD-CP (main-paper Sect. 5) sets a node-wise coefficient from the agreement of the softmax outputs of a node and its neighbours. We re-implement its three variants with fixed published constants, plus our original rule: $\lambda_v = \mathrm{clip}(2\hat{h}_v - 1, 0, 1)$ with soft agreement $\hat{h}_v = d_v^{-1} \sum_{w \in \mathcal{N}(v)} \langle \pi(v), \pi(w) \rangle$.

Label-free, graph-level. We average soft agreement over edges and map it to $\lambda = 0.75\,\mathrm{clip}(\cdot, 0, 1)$ under two normalisations: the analogue of (4) with predicted class shares, and a self-normalised version that divides by the mean self-agreement $\langle \pi(v), \pi(v) \rangle$ minus chance. The constant 0.75 was fixed on Cora and Roman-empire before other graphs were evaluated.

Label-based gate. The same map applied to the adjusted homophily $\hat{a}$ of training edges gives a label gate. HG-DAPS (main-paper Sect. 5) computes training-edge adjusted homophily $\hat{a}$ and sets $\lambda = 0.5\,\mathrm{clip}\big(\hat{a}\, m_{\mathrm{tr}}/(m_{\mathrm{tr}} + 100), 0, 1\big)$, with m_{tr} the number of such edges.

S7 Pseudocode and implementation notes

Algorithm 1 gives the full prediction pipeline for the model-derived SNR-λ selector. The classifier is already fitted. V_{val} is its held-out validation set, and V_{pool} contains the remaining nodes. These sets and $\mathcal{V}_{\text{tr}}$ are disjoint. Assume $K \geq 2$, a nonempty validation set, positive validation true-score variance, and defined training-edge homophily estimates.

Algorithm 1: SNR-based score diffusion and split conformal prediction

Input: Graph $\mathcal{G}$, fitted probabilities $\pi(v)$, training and validation labels, pool V_{pool}, calibration size m, error rate α.
Output: Prediction sets $\mathcal{C}(v)$ for $v \in \mathcal{V}_{\text{test}}$.

Draw independent $U_v \sim \text{Unif}[0,1]$ for all nodes
Compute APS scores S_{vk} using main-paper Eq. (2)
Estimate edge homophily $\hat{h}$ and adjusted homophily $\hat{a}$ on edges with both endpoints in $\mathcal{V}_{\text{tr}}$
$T \leftarrow \{S_{v,y_v} : v \in V_{\text{val}}\}$
$W \leftarrow \{S_{v,k} : v \in V_{\text{val}},\ k \neq y_v\}$
$\hat{\Delta} \leftarrow \text{mean}(W) - \text{mean}(T)$, $\hat{\tau}^2 \leftarrow \text{mean}[(T - \text{mean}(T))^2]$
$\hat{\rho}^2 \leftarrow 1 + \hat{h}(1-\hat{h})\hat{\Delta}^2/\hat{\tau}^2$, $a_+ \leftarrow \max(\hat{a}, 0)$
foreach *node* v **do**
  if $d_v = 0$ **then**
    $\lambda_v \leftarrow 0$, $\tilde{S}_{v,:} \leftarrow S_{v,:}$
  else
    $\lambda_v \leftarrow a_+/(a_+ + \hat{\rho}^2/d_v + 10^{-12})$
    $M_{v,:} \leftarrow d_v^{-1} \sum_{w \in \mathcal{N}(v)} S_{w,:}$
    $\tilde{S}_{v,:} \leftarrow (1-\lambda_v)S_{v,:} + \lambda_v M_{v,:}$
Draw $\mathcal{V}_{\text{cal}} \subset V_{\text{pool}}$ uniformly with $|\mathcal{V}_{\text{cal}}| = m$, $\mathcal{V}_{\text{test}} \leftarrow V_{\text{pool}} \setminus \mathcal{V}_{\text{cal}}$
Reveal calibration labels, set $j \leftarrow \lceil (m+1)(1-\alpha) \rceil$
if $j > m$ **then**
  $\hat{q} \leftarrow +\infty$
else
  $\hat{q} \leftarrow j$-th smallest value in $\{\tilde{S}_{v,y_v} : v \in \mathcal{V}_{\text{cal}}\}$
return $\mathcal{C}(v) = \{k : \tilde{S}_{vk} \leq \hat{q}\}$ *for all* $v \in \mathcal{V}_{\text{test}}$

Implementation details. The score definitions, diffusion rule and validity argument are in main-paper Sects. 3 to 5. Algorithm 1 specifies their execution order and the empirical estimators. Training-edge class shares use degrees within the training-induced subgraph, rather than full-graph degrees. In `src/scores.py`, APS is implemented as the mass preceding the candidate plus $u_v \pi_k(v)$. This equals the manuscript's formula with $U_v = 1 - u_v$. The same random draws are reused across methods in each run. The variance uses divisor $|V_{\text{val}}|$, and the 10^{-12} term matches `snr_lambda`. The stated preconditions exclude undefined homophily or zero-variance estimates. The denominator offset alone does not resolve those cases. `src/conformal.py` implements the order statistic and the $+\infty$ convention.

Computational cost. With n nodes, K classes and $|\mathcal{E}|$ edges, sorting for APS costs $O(nK \log K)$ and diffusion costs $O(|\mathcal{E}|K)$. Scores require $O(nK)$ storage, in addition to the sparse graph. The sparse multiplication computes all neighbour means together. The node loop in Algorithm 1 makes the operations explicit.

S8 Evaluation metrics

The formulas below specify the metrics summarised in main-paper Sect. 6. For one run, let $T = \mathcal{V}_{\text{test}}$, $N = |T|$, $c_v = \mathbb{1}\{y_v \in \mathcal{C}(v)\}$ and $\ell_v = |\mathcal{C}(v)|$. Here $\mathbb{1}\{\cdot\}$ equals one when its condition holds and zero otherwise. Metrics are computed per run before aggregation.

Coverage and set size.

$$\widehat{\text{Cov}} = \frac{1}{N}\sum_{v\in T} c_v, \qquad L = \frac{1}{N}\sum_{v\in T} \ell_v. \tag{S6}$$

Coverage estimates how often the true label is retained. Its target is $1-\alpha$. Mean set size L measures the remaining ambiguity: smaller is better when coverage is comparable. For a method M relative to APS,

$$\text{Change}(M) = 100\left(\frac{\bar{L}_M}{\bar{L}_{\text{APS}}} - 1\right)\%, \tag{S7}$$

where bars denote run means for the specified graph and classifier. Positive values indicate inflation. Negative values indicate shrinkage. This ratio of mean sizes differs from the mean of run-wise ratios used for regret below.

Coverage within groups. For a group $G \subseteq T$, define

$$\widehat{\text{Cov}}(G) = \frac{\sum_{v\in G} c_v}{|G|}, \quad \text{Cov}_{\min} = \min_{G\in\mathcal{G}} \widehat{\text{Cov}}(G), \quad D_{\max} = \max_{G\in\mathcal{G}} |\widehat{\text{Cov}}(G) - (1-\alpha)|. \tag{S8}$$

For worst-class coverage, $\mathcal{G}$ groups nodes by true label. For worst-degree coverage, it contains ten approximately equal-sized groups formed by sorting test-node degrees, with random tie-breaking. The implementation retains groups with at least 20 test nodes. It returns a missing value if none qualify. Thus the reported worst group is the worst among eligible groups. $D_{\max}$, also stored in the artifact, measures absolute deviation, counting both undercoverage and overcoverage. Reported worst-group coverage averages per-run minima. It is not the minimum after pooling runs.

Worst-slice coverage (WSC). A slab selects nodes whose feature embedding z_v projects into an interval:

$$G(a, b, w) = \{v : a \le w^\top z_v \le b\}, \qquad \|w\|_2 = 1. \tag{S9}$$

Randomly divide T into a search half T_A and an evaluation half T_B. Among candidate slabs containing at least 10% of T_A, select

$$G^* \in \arg\min_G \widehat{\mathrm{Cov}}(G \cap T_A), \qquad \widehat{\mathrm{WSC}} = \widehat{\mathrm{Cov}}(G^* \cap T_B). \tag{S10}$$

The experiments search 200 random directions and endpoints on a 20-interval projection-quantile grid. The minimum-size constraint applies to the search half. The held-out slab may contain fewer nodes, and an empty slab gives a missing value. This finite search diagnoses feature-defined subgroups without reporting their optimistically selected search coverage. It is neither an exhaustive worst-subgroup guarantee nor a substitute for class-level coverage.

Oracle regret and aggregation. For cell c (one graph and classifier pair), run r and selector M, let $L_{c,r,M}$ be mean set size. The retrospective scalar-grid reference is

$$L^*_{c,r} = \min_{\lambda \in \{0, 0.1, \ldots, 1\}} L_{c,r,\lambda}, \qquad R_{c,r,M} = 100 \left(\frac{L_{c,r,M}}{L^*_{c,r}} - 1 \right). \tag{S11}$$

Regret is expressed in percent. With J_c paired runs in cell c, set $\bar{R}_{c,M} = J_c^{-1} \sum_r R_{c,r,M}$. Over the C evaluated cells,

$$R_{\mathrm{mean},M} = \frac{1}{C} \sum_c \bar{R}_{c,M}, \; R_{\mathrm{worst},M} = \max_c \bar{R}_{c,M}, \; P_{2,M} = \frac{100}{C} \sum_c \mathbb{1}\{\bar{R}_{c,M} \le 2\}. \tag{S12}$$

Cells have equal weight. Homophilous and heterophilous means restrict the same average to their respective cells. "Worst" means the worst cell mean, not the worst run. $P_{2,M}$ is the percentage of cells within 2% of the reference. Negative regret is possible because node-wise or signed rules lie outside the oracle's non-negative scalar grid. The random-grid baseline uses $11^{-1} \sum_\lambda L_{c,r,\lambda}$, its exact expected size under a uniform grid choice, before computing regret.

Additional stored diagnostics. Singleton rate is $N^{-1} \sum_v \mathbb{1}\{\ell_v = 1\}$. Empty-set rate is $N^{-1} \sum_v \mathbb{1}\{\ell_v = 0\}$. Size-stratified coverage violation is $\max_B |\widehat{\mathrm{Cov}}(B) - (1-\alpha)|$, using nonempty-size bins $\{1\}$, $\{2,3\}$, $\{4,5,6\}$, $\{7,\ldots,10\}$, $\{11,\ldots,20\}$, $\{21,\ldots,50\}$ and $\{51,\ldots,K\}$, restricted to feasible sizes and bins with at least 20 nodes. The implementation returns zero when no bin qualifies. This convention is not evidence of perfect conditional coverage. Classifier accuracy is $N^{-1} \sum_v \mathbb{1}\{\arg\max_k \pi_k(v) = y_v\}$, which measures single-label prediction rather than conformal coverage. For calibration weights w_i, the shift diagnostic reports effective sample size $\mathrm{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$. Small values indicate that a few nodes dominate the weighted calibration sample.

Uncertainty summaries. For a reported mean $\bar{x}$, each bootstrap replicate resamples the relevant units with replacement and recomputes the mean. The 2.5th and 97.5th percentiles of 1000 replicates define a 95% percentile interval. Tables showing $\bar{x} \pm h$ use half its width, $h = (q_{0.975} - q_{0.025})/2$. Runs are the units for

run-level summaries. Cells are the units for the regret-mean interval, which is descriptive because classifiers share graphs. Crossover intervals instead resample graph seeds, as described in main-paper Sect. 7.1.